

WASP-43b

R-0462

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_240129 · created 2026-07-19T21:11:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460338.9429 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.043
Transit depth (Rp/Rs)^2	1.54 +/- 1.06 [%]
Orbital Inclination (inc)	83.54 +/- 2.77
Airmass coefficient 1 (a1)	1.225 +/- 0.033
Airmass coefficient 2 (a2)	-0.131 +/- 0.021
Scatter in the residuals of the lightcurve fit is	2.71 %
Best Comparison Star	#2 - [396, 44]
Optimal Method	PSF photometry
Transit Duration (day)	0.0505 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.124 ± 0.043 vs 0.1594 (deviation 0.67σ)
Duration fit vs published	0.0505 d vs 0.0512 d (deviation 0.06σ)
Mid-time O–C	3.0 min
Depth SNR	2.4
Residual scatter	2.71 %

Light curve

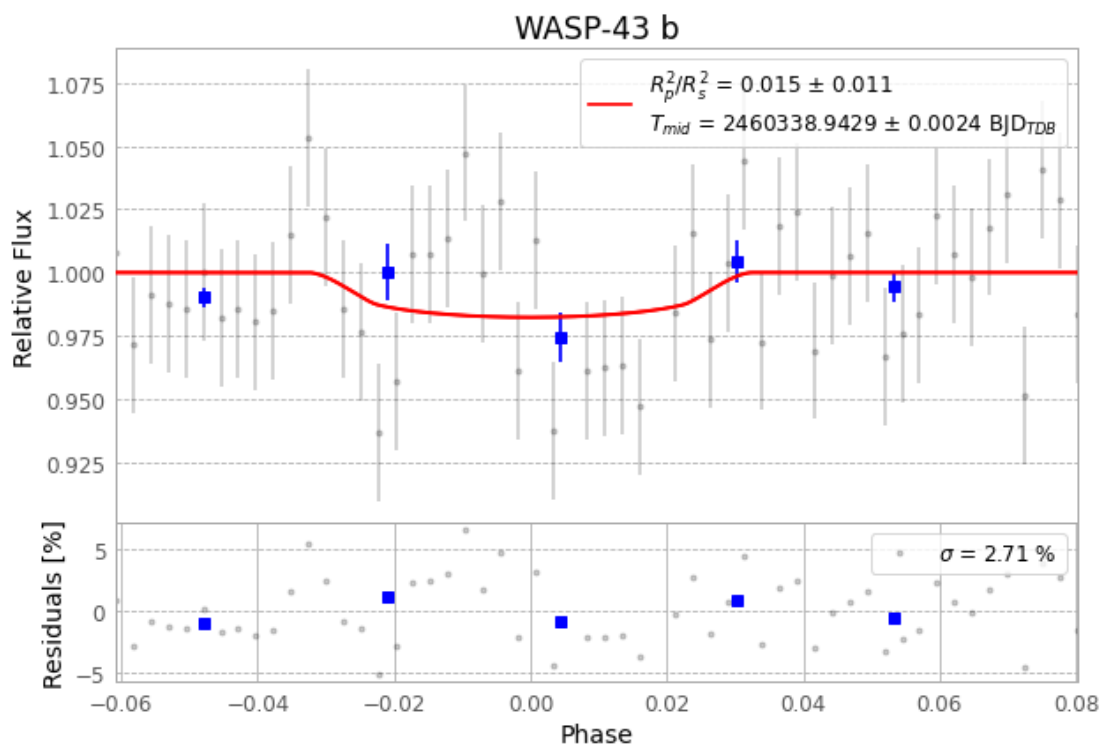


Figure 1 — FinalLightCurve_WASP-43 b_2024-01-29.png (SHA-256 729a7d5e7e5d1197...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [484, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4768698432582e2e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

56 FITS frames (37 MB), WASP-43240129091815.FITS → WASP-43240129120316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240129.log (fff97157be23...), FinalLightCurve_WASP-43 b_2024-01-29.png (729a7d5e7e5d...), FinalLightCurve_WASP-43 b_2024-01-29.csv (fea7b5f1d214...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 21:11Z.